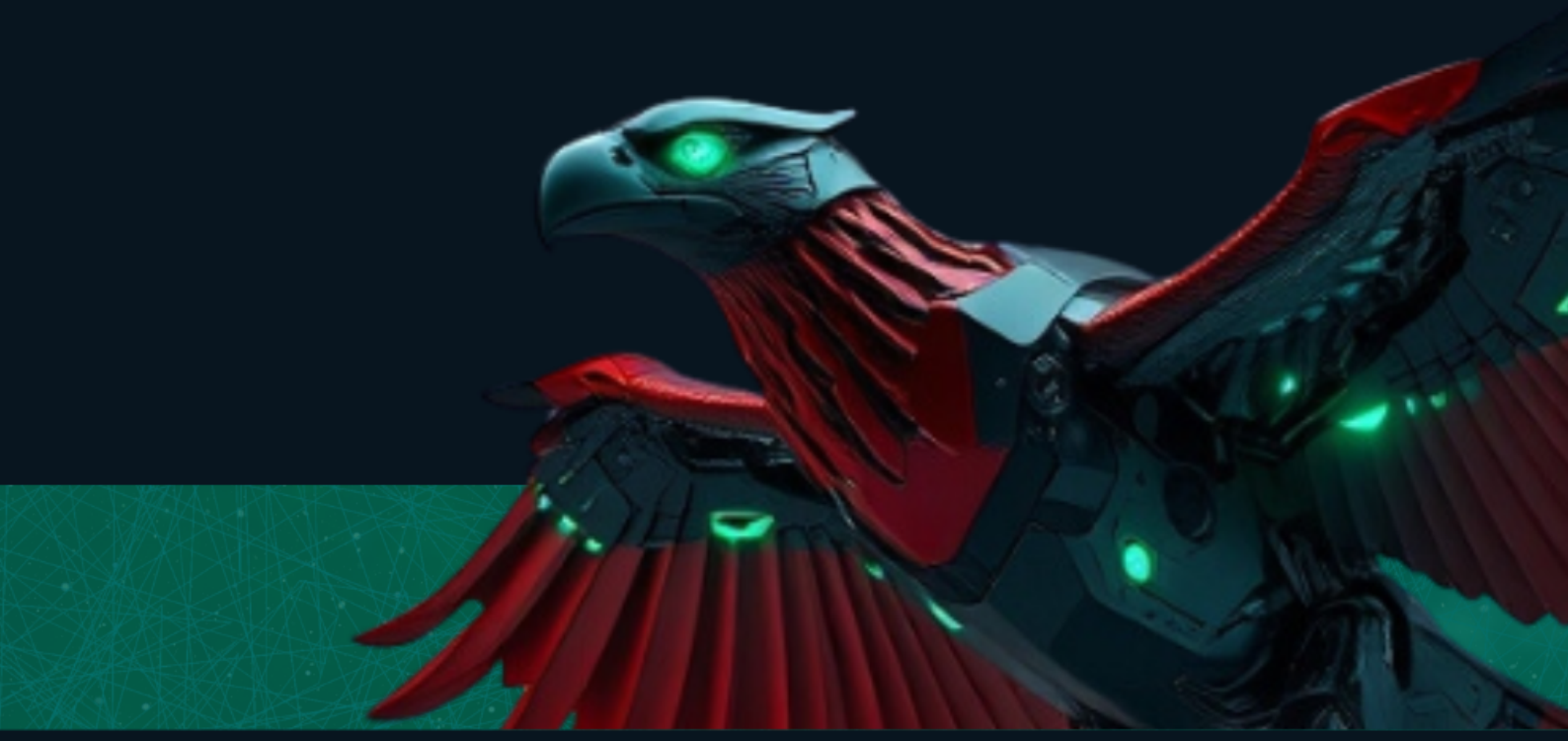
SOCBYTE
ALL BYTES SECURED

CASE REPORT



Case Details

Reference ID	CTHTY-172
Title	Incident Response: gsecdump
Status	Closed
Severity	High
Tactics	Credential Access
Created By	christopher.mirro@thevsoc.com
Created At	May 20, 2026, 08:58 PM
Category	Malware
Sub-Category	Virus
Tenant	Cythority
Age	22 hours
Offense ID	—

Groups & Contributors

Assigned Groups

Security Engineering-Cythority

SOC-Cythority

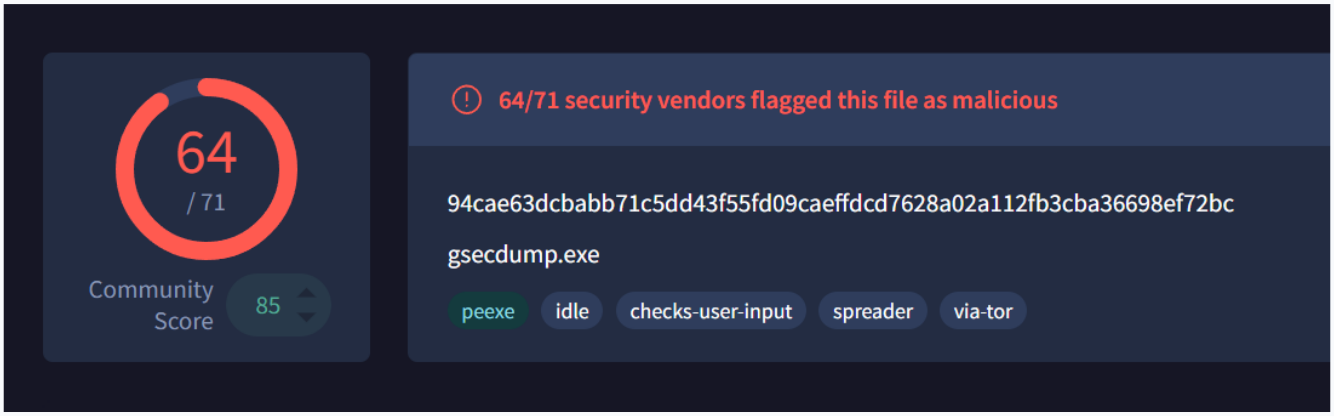
Analysis Summary

On May 20th, 2026 at 8:13 PM , Cythrority's VSOC investigated a powershell.exe (interactive session) alert/detection on SentinelOne. Gsecdump is a well-known, open-source credential dumping tool. It is primarily used by security auditors (and threat actors) to extract user account passwords and password hashes from the Windows Local Security Authority Subsystem Service (LSASS) memory, as well as secrets from the domain cached credentials. The execution of gsecdump.exe from the Atomic Red Team directory triggered critical indicators across multiple MITRE ATT&CK tactics, primarily centered around Credential Access. The endpoint agent successfully detected and intervened against OS Credential Dumping, specifically blocking multiple invasive attempts to open and read sensitive authentication data directly from the LSASS Memory and targeted password stores containing Cached Domain Credentials. This malicious payload was initiated via an interactive PowerShell session using indirect command-line arguments to evade standard execution paths. Static and dynamic analysis engines also flagged signs of Defense Evasion, noting that the process was highly suspicious due to an embedded PE file resource characteristic of Obfuscated Files or Information. While the runtime protection successfully mitigated the core threat by blocking LSASS access, the alert correlates these actions to severe downstream risks; had the credential harvesting succeeded, a threat actor could leverage compromised Valid Accounts and exploit established Trusted

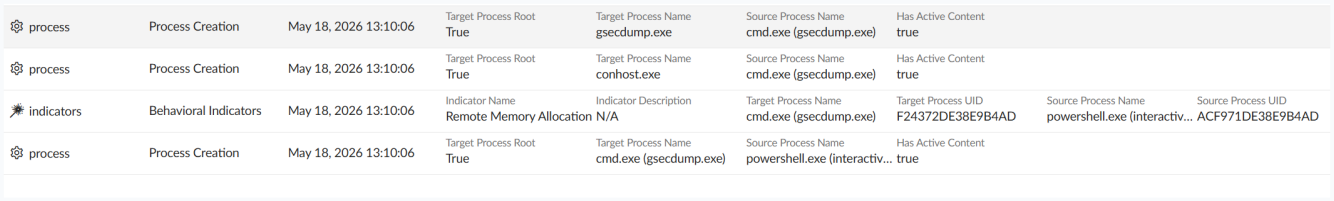
Relationships to achieve Privilege Escalation, establish Persistence, and move laterally across the organizational infrastructure. In regards to the file hash, when put into VirusTotal, it came up as highly malicious.

Upon further investigation, an analysis of the alert indicators points to a combination of highly malicious technical signatures executed within a strictly suspicious contextual environment, ultimately leading to a conclusion of an authorized security simulation. The behavior of the binary itself is fundamentally malicious; both static and dynamic engine rules flagged the process for executing invasive attempts to open and read sensitive authentication data directly from the Windows Local Security Authority Subsystem Service (LSASS). Furthermore, the file triggered critical defense evasion indicators because the process is suspicious as it is packed and contains an embedded PE file resource. Common technical methods used to obfuscate code and bypass traditional static signature detection.

Evidence



Evidence 1



process	Process Creation	May 18, 2026 13:10:06	Target Process Root: True	Target Process Name: gsecdump.exe	Source Process Name: cmd.exe (gsecdump.exe)	Has Active Content: true			
process	Process Creation	May 18, 2026 13:10:06	Target Process Root: True	Target Process Name: conhost.exe	Source Process Name: cmd.exe (gsecdump.exe)	Has Active Content: true			
indicators	Behavioral Indicators	May 18, 2026 13:10:06	Indicator Name: Remote Memory Allocation	Indicator Description: N/A	Target Process Name: cmd.exe (gsecdump.exe)	Target Process UID: F24372DE38E9B4AD	Source Process Name: powershell.exe (interactiv...	Source Process UID: ACF971DE38E9B4AD	
process	Process Creation	May 18, 2026 13:10:06	Target Process Root: True	Target Process Name: cmd.exe (gsecdump.exe)	Source Process Name: powershell.exe (interactiv...	Has Active Content: true			

Evidence 2

Remediation

- Verify Test Authorization
- Isolate Endpoint if Unauthorized
- Update Threat Status and Verdict
- Perform a Full Disk Scan
- Review Local Administrator Rights
- Validate LSASS Protection Policies